🌐 Networking Fundamentals Overview

This note synthesizes key concepts from Cisco networking modules, covering switching fundamentals, router and switch configuration, address resolution protocols, and DHCP for both IPv4 and IPv6.

📚 Content Sources

- Section 1: Introduces **switching concepts** and MAC address table fundamentals using animations and videos from a 20-page PDF document. (source)
- Section 2: Guides through **router configuration** steps, including interface setup and default-gateway assignment from a 20-page PDF document. (source)
- Section 3: Covers **switch configuration** and initial **end-device** settings such as passwords and IP addressing from a 44-page PDF document. (source)
- Section 4: Explains **ARP** and **IPv6 Neighbor Discovery** mechanisms for mapping IP to MAC addresses from a 12-page PDF document. (source)
- Section 5: Details **DHCPv4** and **DHCPv6** operation, configuration, and troubleshooting from an 18-page PDF document. (source)

Section 1: Switching Concepts and Frame Forwarding 🔄

This section covers Layer 2 switching fundamentals and frame forwarding mechanisms from a 20-page PDF document (source)

Module Overview

The module blends **theory** (animations, videos) with **practice** (PT activities, labs) and **assessment** (CYU, quizzes) to deliver comprehensive Layer 2 switching knowledge.

Core Objectives

- **Frame Forwarding**: Explain how frames are forwarded in a switched network
- **Switching Domains**: Compare collision domains to broadcast domains

Frame Forwarding Fundamentals

Key Terminology

- **Ingress**: Interface where a frame **enters** the switch
- **Egress**: Interface where a frame **exits** the switch
- **Critical Rule**: A switch never forwards traffic back out the same interface it arrived on

MAC Address Table (CAM Table)

The switch maintains a **Content-Addressable Memory (CAM) table** that maps **source MAC addresses** to **ports**

Learn-and-Forward Process

1. **Learn**: Examine source MAC address
   - If **new**: Add to CAM table with ingress port
   - If **exists**: Reset timeout (default 5 minutes)
2. **Forward**: Examine destination MAC address
   - If **exists** in CAM table: Forward out associated egress port
   - If **not exists**: **Flood** out all interfaces except ingress port

Switching Forwarding Methods

Method	Description	Use Case
Store-and-Forward	Receives entire frame, checks FCS for CRC errors, buffers, then forwards	Cisco preferred; error-free forwarding
Cut-Through	Begins forwarding after destination MAC identified (after 6 bytes)	Low-latency environments (≤10μs)

Method Characteristics

Store-and-Forward:

- ✅ Error checking (discards bad FCS)
- ✅ Buffering for speed mismatches

Cut-Through:

- ✅ Minimal latency
- ❌ No FCS check (errors may propagate)

- ❌ Cannot bridge different speeds

Switching Domains

Collision Domains

- **Full-duplex links**: **Eliminate** collision domains
- **Half-duplex links**: **Create** collision domains
- **Auto-negotiation**: Determines optimal speed/duplex settings

Broadcast Domains

- Includes all devices receiving **Layer 2 broadcast frames**
- Spans all **Layer 1** and **Layer 2** devices
- Only **Layer 3 device (router)** can break broadcast domain
- Switch **floods** broadcasts out all ports except ingress port

Congestion Mitigation Features

- Fast port speeds (up to 100 Gbps)
- Fast internal switching
- Large frame buffers
- High port density

Section 2: Basic Router Configuration 🔧

This section covers initial router configuration including settings, interfaces, and default gateway configuration from a 20-page PDF document (source)

Module Objectives

- Configure initial settings on IOS Cisco router
- Configure two active interfaces on Cisco IOS router
- Configure devices to use default gateway

Initial Router Settings Configuration

Basic Configuration Steps

Step	Command	Purpose
1	hostname	Assign device name

2	enable secret	Secure privileged EXEC mode
3	line console 0 → password → login	Secure console access
4	line vty 0 4 → password → login	Secure remote access
5	transport input {ssh	telnet}`
6	service password encryption	Encrypt passwords
7	banner motd ##	Legal notification
8	copy running-config startup-config	Save to NVRAM

Interface Configuration

Configuration Syntax

```
Router(config)# interface
Router(config-if)# description
Router(config-if)# ip address
Router(config-if)# ipv6 address /
Router(config-if)# no shutdown
```

Verification Commands

- show ip interface brief - IPv4 status
- show ipv6 interface brief - IPv6 status
- show ip route - IPv4 routing table
- show ipv6 route - IPv6 routing table
- show interfaces - Detailed statistics

Default Gateway Configuration

- **Host default gateway**: Router interface on same subnet
- **Switch default gateway**: ip default-gateway for remote management

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Section 3: Basic Switch and End-Device Configuration 💻

This section covers switch configuration, IOS navigation, and end-device setup from a 44-page PDF document ([source]())

Module Objectives

- Access Cisco IOS devices
- Navigate IOS command modes
- Configure basic device settings
- Configure IP addressing on hosts

IOS Access Methods

Method	Description
Console	Physical management port
SSH	Secure remote access (recommended)
Telnet	Insecure remote access

IOS Navigation Modes

Mode	Prompt	Access Method
User EXEC	>	Initial access
Privileged EXEC	#	enable command
Global Config	(config)#	configure terminal
Line Config	(config-line)#	line console 0 or line vty 0 15
Interface Config	(config-if)#	interface

Command Structure

- **Keywords**: Pre-defined terms (ip, protocols)
- **Arguments**: User-supplied values (IP addresses)
- **Help**: ? for context-sensitive help
- **Shortcuts**: Tab completion, Ctrl+C, Ctrl+Z

Basic Device Configuration

Essential Settings

1. **Hostname**: hostname (≤64 chars, no spaces)
2. **Passwords**:
    - Console: line console 0 → password → login
    - Privileged: enable secret
    - VTY: line vty 0 15 → password → login
3. **Encryption**: service password-encryption
4. **Banner**: banner motd ##

Configuration Files

- **Startup-config**: NVRAM (persistent)
- **Running-config**: RAM (active)
- **Save**: copy running-config startup-config
- **Clear**: erase startup-config → reload

IP Addressing Configuration

Manual Configuration

- Windows: Control Panel → Network Settings
- Requires: IP address, subnet mask, default gateway, DNS

DHCP Configuration

- Automatic IP assignment
- DHCPv4/v6 support
- Default on most systems

Switch Virtual Interface (SVI)

```
configure terminal
interface vlan 1
ip address
no shutdown
```

Section 4: Address Resolution and ARP Protocol 📡

This section examines MAC-IP relationships and ARP functionality from a 12-page PDF document (source)

Address Types Comparison

Layer	Address	Purpose
Layer 2	**MAC Address**	48-bit hardware identifier for local delivery
Layer 3	**IP Address**	Logical address for cross-network routing

Communication Scenarios

Same Network Communication

- Destination MAC = Destination host's MAC
- ARP resolves IP to MAC if unknown

Remote Network Communication

- Destination MAC = Default gateway's MAC
- ARP resolves gateway's IP to MAC

ARP (Address Resolution Protocol)

Core Functions

1. **Resolution**: IPv4 → MAC mapping
2. **Cache Maintenance**: Stores mappings in ARP table

ARP Process

1. Check ARP table for destination IP
2. If no entry exists, broadcast ARP request
3. Target device replies with ARP response
4. Update ARP table with new mapping

ARP Table Lifecycle

- **Timer expiration**: Entry removed after timeout (typically 60s)
- **Manual removal**: clear ip arp (Cisco) or arp -d (Windows)
- **Entry refresh**: New ARP request/reply updates timer

Viewing ARP Tables

Cisco IOS:

```
show ip arp
```

Windows:

```
arp -a
```

ARP Security Issues

- **Broadcast storms**: Excessive ARP requests consume bandwidth
- **ARP spoofing**: Forged ARP replies enable man-in-the-middle attacks
- **Mitigation**: Dynamic ARP Inspection (DAI), port security

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Section 5: DHCP Configuration and Operation 🌐

This section covers DHCPv4 and DHCPv6 configuration, operation, and troubleshooting from an 18-page PDF document (source)

DHCP Overview

Dynamic Host Configuration Protocol automates IPv4/IPv6 configuration assignment, eliminating manual configuration needs.

DHCPv4 Address Allocation Methods

Method	Description	Use Case
Manual	Administrator pre-assigns specific IP	Servers, printers
Automatic	Server permanently binds address to client	Rarely used
Dynamic	Server leases address for time period	Most common for hosts

DHCPv4 Message Exchange

1. **DHCPDISCOVER**: Client broadcasts request
2. **DHCPOFFER**: Server offers IP configuration
3. **DHCPREQUEST**: Client requests offered IP
4. **DHCPACK**: Server acknowledges lease

Cisco IOS DHCP Server Configuration

Configuration Steps

1. **Exclude addresses**: ip dhcp excluded-address
2. **Create pool**: ip dhcp pool
3. **Define network**: network
4. **Set gateway**: default-router
5. **Configure DNS**: dns-server
6. **Domain name**: domain-name

Verification Commands

- show ip dhcp binding - Current client leases
- show ip dhcp server statistics - DHCP traffic stats
- show running-config | section dhcp - DHCP configuration

DHCP Relay

- **IP Helper**: ip helper-address
- Forwards DHCP broadcasts between subnets
- Enables centralized DHCP services

DHCP Client Configuration

```
interface GigabitEthernet0/0
 ip address dhcp
 no shutdown
```

DHCPv6 Overview

- **Stateless**: Provides config info only (DNS, domain)
- **Stateful**: Full address assignment
- Uses **DHCPv6** or **SLAAC** for dynamic allocation

Troubleshooting Commands

- debug ip dhcp server events - Monitor DHCP activity

- debug ip dhcp server packet - Packet-level details
- **Caution**: Debugging generates significant output

🎯 Putting it All Together

📡 How Layer 2 and Layer 3 Interact

Layer	Primary Role	Key Mechanism
L2 (Switching)	Moves **frames** inside a broadcast domain.	Learn-and-forward using the **CAM table** (source MAC → port).
L3 (Routing)	Moves **packets** between broadcast domains.	**Default gateway** (router) forwards traffic to remote subnets.

> *Ingress* – the port where a frame first arrives.
> *Egress* – the port a frame exits after lookup.

Switches never send a frame back out the same ingress port; they either forward to a single egress port (known MAC) or **flood** to all other ports (unknown MAC or broadcast).

🔀 Forwarding Styles on Modern Switches

Method	When it's used	Main trade-off
Store-and-Forward	All traffic by default.	Checks the **FCS**, discarding corrupted frames – adds a few µs latency.
Cut-Through	Low-latency environments.	Starts forwarding after reading the destination MAC, so errors can slip through.

🔐 Securing Devices from the First Command

- **Hostname** gives each device a unique identity.
- **enable secret** protects privileged mode with an encrypted password.
- **Console & VTY passwords** (line console, line vty) lock local and remote access.

- **service password-encryption** hashes all clear-text passwords.
- **MOTD banner** warns unauthorized users.

> *A good password* mixes upper-/lower-case letters, numbers, and symbols, and exceeds eight characters.

🌐 IP Addressing Basics

Address type	Layer	Use
MAC	2	NIC-to-NIC delivery inside the same LAN.
IPv4 / IPv6	3	Global routing across subnets and the Internet.

When a host's destination IP is **local**, it resolves the **destination MAC** via **ARP** (or IPv6 Neighbor Discovery). If the destination is **remote**, it resolves the **router's MAC** instead and sends the frame to that gateway.

📡 ARP & Neighbor Discovery in Action

1. Host checks its **ARP cache** for the needed IPv4-→-MAC mapping.
2. **Cache miss** triggers a **broadcast ARP request**: "Who has X? Tell Y."
3. Owner replies with an **ARP reply**, and the entry is cached for a limited time (often ~60 s).
4. IPv6 uses **ICMPv6 Neighbor Discovery** to perform the same function without ARP.

> *ARP spoofing* injects false replies, enabling man-in-the-middle attacks; mitigation includes **Dynamic ARP Inspection** and **port security**.

▯ DHCP – Automating Address Assignment

Allocation	How it works	Typical scenario
Manual (reservation)	Server always hands the same IP to a specific MAC.	Servers, printers.
Automatic	Server assigns an IP permanently; the lease never expires.	Rare, static-like use.

Dynamic	Server leases an IP for a configurable period; client must renew.	Everyday workstations, laptops.

Four-message handshake:

1. **DHCPDISCOVER** – client broadcasts (source IP 0.0.0.0).
2. **DHCPOFFER** – server proposes an address and options.
3. **DHCPREQUEST** – client selects an offer.
4. **DHCPACK** – server finalises the lease.

A Cisco router can act as:

- **DHCP server** – define excluded addresses, a pool (network, default-router, dns-server, domain-name).
- **DHCP client** – ip address dhcp on the interface.
- **Relay agent** – ip helper-address to forward broadcasts across subnets.

🛠️ Verification & Troubleshooting

Tool	What it reveals
show ip dhcp binding	Current IP-to-MAC leases.
show ip dhcp pool	Free vs. used addresses in each pool.
s̀how running-config	section dhcp`
debug ip dhcp server events	Live DHCP traffic flow (use sparingly).
show ip arp / arp -a	IPv4 MAC tables on routers and PCs.
show vlan / show mac address-table	Switch learning state.

If a host cannot reach a remote network:

1. Verify **ARP cache** for the correct gateway MAC.
2. Check the **default gateway IP** on the host.
3. Confirm the **router interface** is up (no shutdown).
4. Ensure DHCP (if used) has supplied the correct **subnet mask** and **gateway**.

📈 Putting the Pieces Together

- **Switches** learn MACs, forward frames, and isolate **collision domains** (full-duplex eliminates collisions).

- **Routers** provide the **default gateway**, bridging broadcast domains and enabling inter-subnet communication.
- **Device configuration** (hostname, passwords, interface IPs, SVI) establishes a secure, manageable foundation.
- **ARP/Neighbor Discovery** tie IPv4/IPv6 addresses to MACs, allowing switches to deliver frames correctly.
- **DHCP** automates the distribution of those IP settings, while relay agents extend the service across multiple LAN segments.

Together, these mechanisms form the backbone of a modern enterprise LAN: **Layer 2** handles rapid local delivery, **Layer 3** routes between networks, and **automation/security** (DHCP, passwords, ARP protection) keep the environment both efficient and safe.

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Section 1: Switching Concepts and Frame Forwarding 🔄

This section covers Layer 2 switching fundamentals and frame forwarding mechanisms from a 20-page PDF document (source)

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Core Objectives

- **Frame Forwarding**: Explain how frames are forwarded in a switched network
- **Switching Domains**: Compare collision domains to broadcast domains

Frame Forwarding Fundamentals

Key Terminology

- **Ingress**: Interface where a frame **enters** the switch
- **Egress**: Interface where a frame **exits** the switch
- **Critical Rule**: A switch never forwards traffic back out the same interface it arrived on

MAC Address Table (CAM Table)

The switch maintains a **Content-Addressable Memory (CAM) table** that maps **source MAC addresses** to **ports**

Learn-and-Forward Process

1. **Learn**: Examine source MAC address
    - If **new**: Add to CAM table with ingress port
    - If **exists**: Reset timeout (default 5 minutes)
2. **Forward**: Examine destination MAC address
    - If **exists** in CAM table: Forward out associated egress port
    - If **not exists**: **Flood** out all interfaces except ingress port

Switching Forwarding Methods

Method	Description	Use Case
Store-and-Forward	Receives entire frame, checks FCS for CRC errors,	Cisco preferred; error-free forwarding

	buffers, then forwards	
Cut-Through	Begins forwarding after destination MAC identified (after 6 bytes)	Low-latency environments (≤10μs)

Method Characteristics

Store-and-Forward:

- ✅ Error checking (discards bad FCS)
- ✅ Buffering for speed mismatches

Cut-Through:

- ✅ Minimal latency
- ❌ No FCS check (errors may propagate)
- ❌ Cannot bridge different speeds

Switching Domains

Collision Domains

- **Full-duplex links**: **Eliminate** collision domains
- **Half-duplex links**: **Create** collision domains
- **Auto-negotiation**: Determines optimal speed/duplex settings

Broadcast Domains

- Includes all devices receiving **Layer 2 broadcast frames**
- Spans all **Layer 1** and **Layer 2** devices
- Only **Layer 3 device (router)** can break broadcast domain
- Switch **floods** broadcasts out all ports except ingress port

Congestion Mitigation Features

- Fast port speeds (up to 100 Gbps)
- Fast internal switching
- Large frame buffers
- High port density

Section 2: Basic Router Configuration 🔧

This section covers initial router configuration including settings, interfaces, and default gateway configuration from a 20-page PDF document (source)

Module Objectives

- Configure initial settings on IOS Cisco router
- Configure two active interfaces on Cisco IOS router
- Configure devices to use default gateway

Initial Router Settings Configuration

Basic Configuration Steps

Step	Command	Purpose
1	hostname	Assign device name
2	enable secret	Secure privileged EXEC mode
3	line console 0 → password → login	Secure console access
4	line vty 0 4 → password → login	Secure remote access
5	transport input {ssh	telnet}`
6	service password encryption	Encrypt passwords
7	banner motd ##	Legal notification
8	copy running-config startup-config	Save to NVRAM

Interface Configuration

Configuration Syntax

```
Router(config)# interface
Router(config-if)# description
Router(config-if)# ip address
```

```
Router(config-if)# ipv6 address /
Router(config-if)# no shutdown
```

Verification Commands

- show ip interface brief - IPv4 status
- show ipv6 interface brief - IPv6 status
- show ip route - IPv4 routing table
- show ipv6 route - IPv6 routing table
- show interfaces - Detailed statistics

Default Gateway Configuration

- **Host default gateway**: Router interface on same subnet
- **Switch default gateway**: ip default-gateway for remote management

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Section 3: Basic Switch and End-Device Configuration 💻

This section covers switch configuration, IOS navigation, and end-device setup from a 44-page PDF document ([source]())

Module Objectives

- Access Cisco IOS devices
- Navigate IOS command modes
- Configure basic device settings
- Configure IP addressing on hosts

IOS Access Methods

Method	Description
Console	Physical management port
SSH	Secure remote access (recommended)
Telnet	Insecure remote access

IOS Navigation Modes

Mode	Prompt	Access Method

User EXEC	>	Initial access
Privileged EXEC	#	enable command
Global Config	(config)#	configure terminal
Line Config	(config-line)#	line console 0 or line vty 0 15
Interface Config	(config-if)#	interface

Command Structure

- **Keywords**: Pre-defined terms (ip, protocols)
- **Arguments**: User-supplied values (IP addresses)
- **Help**: ? for context-sensitive help
- **Shortcuts**: Tab completion, Ctrl+C, Ctrl+Z

Basic Device Configuration

Essential Settings

1. **Hostname**: hostname (≤64 chars, no spaces)
2. **Passwords**:
   - Console: line console 0 → password → login
   - Privileged: enable secret
   - VTY: line vty 0 15 → password → login
3. **Encryption**: service password-encryption
4. **Banner**: banner motd ##

Configuration Files

- **Startup-config**: NVRAM (persistent)
- **Running-config**: RAM (active)
- **Save**: copy running-config startup-config
- **Clear**: erase startup-config → reload

IP Addressing Configuration

Manual Configuration

- Windows: Control Panel → Network Settings
- Requires: IP address, subnet mask, default gateway, DNS

DHCP Configuration

- Automatic IP assignment
- DHCPv4/v6 support
- Default on most systems

Switch Virtual Interface (SVI)

```
configure terminal
interface vlan 1
ip address
no shutdown
```

Section 4: Address Resolution and ARP Protocol 📡

This section examines MAC-IP relationships and ARP functionality from a 12-page PDF document ([source]())

Address Types Comparison

Layer	Address	Purpose
Layer 2	**MAC Address**	48-bit hardware identifier for local delivery
Layer 3	**IP Address**	Logical address for cross-network routing

Communication Scenarios

Same Network Communication

- Destination MAC = Destination host's MAC
- ARP resolves IP to MAC if unknown

Remote Network Communication

- Destination MAC = Default gateway's MAC
- ARP resolves gateway's IP to MAC

ARP (Address Resolution Protocol)

Core Functions

1. **Resolution**: IPv4 → MAC mapping
2. **Cache Maintenance**: Stores mappings in ARP table

ARP Process

1. Check ARP table for destination IP
2. If no entry exists, broadcast ARP request
3. Target device replies with ARP response
4. Update ARP table with new mapping

ARP Table Lifecycle

- **Timer expiration**: Entry removed after timeout (typically 60s)
- **Manual removal**: clear ip arp (Cisco) or arp -d (Windows)
- **Entry refresh**: New ARP request/reply updates timer

Viewing ARP Tables

Cisco IOS:

```
show ip arp
```

Windows:

```
arp -a
```

ARP Security Issues

- **Broadcast storms**: Excessive ARP requests consume bandwidth
- **ARP spoofing**: Forged ARP replies enable man-in-the-middle attacks
- **Mitigation**: Dynamic ARP Inspection (DAI), port security

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Section 5: DHCP Configuration and Operation 🌐

This section covers DHCPv4 and DHCPv6 configuration, operation, and troubleshooting from an 18-page PDF document (source)

DHCP Overview

Dynamic Host Configuration Protocol automates IPv4/IPv6 configuration assignment, eliminating manual configuration needs.

DHCPv4 Address Allocation Methods

Method	Description	Use Case
Manual	Administrator pre-assigns specific IP	Servers, printers
Automatic	Server permanently binds address to client	Rarely used
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DHCPv4 Message Exchange

1. **DHCPDISCOVER**: Client broadcasts request
2. **DHCPOFFER**: Server offers IP configuration
3. **DHCPREQUEST**: Client requests offered IP
4. **DHCPACK**: Server acknowledges lease

Cisco IOS DHCP Server Configuration

Configuration Steps

1. **Exclude addresses**: ip dhcp excluded-address
2. **Create pool**: ip dhcp pool
3. **Define network**: network
4. **Set gateway**: default-router
5. **Configure DNS**: dns-server
6. **Domain name**: domain-name

Verification Commands

- show ip dhcp binding - Current client leases
- show ip dhcp server statistics - DHCP traffic stats
- show running-config | section dhcp - DHCP configuration

DHCP Relay

- **IP Helper**: ip helper-address
- Forwards DHCP broadcasts between subnets
- Enables centralized DHCP services

DHCP Client Configuration

```
interface GigabitEthernet0/0
 ip address dhcp
 no shutdown
```

DHCPv6 Overview

- **Stateless**: Provides config info only (DNS, domain)
- **Stateful**: Full address assignment
- Uses **DHCPv6** or **SLAAC** for dynamic allocation

Troubleshooting Commands

- debug ip dhcp server events - Monitor DHCP activity
- debug ip dhcp server packet - Packet-level details
- **Caution**: Debugging generates significant output

🎯 Putting it All Together

📡 How Layer 2 and Layer 3 Interact

Layer	Primary Role	Key Mechanism
L2 (Switching)	Moves **frames** inside a broadcast domain.	Learn-and-forward using the **CAM table** (source MAC → port).
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🔀 Forwarding Styles on Modern Switches

Method	When it's used	Main trade-off
Store-and-Forward	All traffic by default.	Checks the **FCS**, discarding corrupted frames – adds a few µs latency.
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> *A good password* mixes upper-/lower-case letters, numbers, and symbols, and exceeds eight characters.

🌐 IP Addressing Basics

Address type	Layer	Use
MAC	2	NIC-to-NIC delivery inside the same LAN.
IPv4 / IPv6	3	Global routing across subnets and the Internet.

When a host's destination IP is **local**, it resolves the **destination MAC** via **ARP** (or IPv6 Neighbor Discovery). If the destination is **remote**, it resolves the **router's MAC** instead and sends the frame to that gateway.

📡 ARP & Neighbor Discovery in Action

1. Host checks its **ARP cache** for the needed IPv4-→-MAC mapping.
2. **Cache miss** triggers a **broadcast ARP request**: "Who has X? Tell Y."
3. Owner replies with an **ARP reply**, and the entry is cached for a limited time (often ~60 s).

4. IPv6 uses **ICMPv6 Neighbor Discovery** to perform the same function without ARP.

> *ARP spoofing* injects false replies, enabling man-in-the-middle attacks; mitigation includes **Dynamic ARP Inspection** and **port security**.

▯ DHCP – Automating Address Assignment

Allocation	How it works	Typical scenario
Manual (reservation)	Server always hands the same IP to a specific MAC.	Servers, printers.
Automatic	Server assigns an IP permanently; the lease never expires.	Rare, static-like use.
Dynamic	Server leases an IP for a configurable period; client must renew.	Everyday workstations, laptops.

Four-message handshake:

1. **DHCPDISCOVER** – client broadcasts (source IP 0.0.0.0).
2. **DHCPOFFER** – server proposes an address and options.
3. **DHCPREQUEST** – client selects an offer.
4. **DHCPACK** – server finalises the lease.

A Cisco router can act as:

- **DHCP server** – define excluded addresses, a pool (network, default-router, dns-server, domain-name).
- **DHCP client** – ip address dhcp on the interface.
- **Relay agent** – ip helper-address to forward broadcasts across subnets.

🛠️ Verification & Troubleshooting

Tool	What it reveals
show ip dhcp binding	Current IP-to-MAC leases.
show ip dhcp pool	Free vs. used addresses in each pool.
`show running-config	section dhcp`

debug ip dhcp server events	Live DHCP traffic flow (use sparingly).
show ip arp / arp -a	IPv4 MAC tables on routers and PCs.
show vlan / show mac address-table	Switch learning state.

If a host cannot reach a remote network:

1. Verify **ARP cache** for the correct gateway MAC.
2. Check the **default gateway IP** on the host.
3. Confirm the **router interface** is up (no shutdown).
4. Ensure DHCP (if used) has supplied the correct **subnet mask** and **gateway**.

📈 Putting the Pieces Together

- **Switches** learn MACs, forward frames, and isolate **collision domains** (full-duplex eliminates collisions).
- **Routers** provide the **default gateway**, bridging broadcast domains and enabling inter-subnet communication.
- **Device configuration** (hostname, passwords, interface IPs, SVI) establishes a secure, manageable foundation.
- **ARP/Neighbor Discovery** tie IPv4/IPv6 addresses to MACs, allowing switches to deliver frames correctly.
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